

Aryan Bhagat

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EDUCATION

California State University, East Bay | *M.S. in Computer Science* — GPA: 3.9/4.0 Aug 2024 – May 2026
Gujarat Technological University | *B.E. in Computer Engineering* — GPA: 8.8/10.0 Oct 2020 – May 2024

SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript

AI/ML: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, OpenCV, YOLO

LLM Engineering: LangChain, LangGraph, RAG, Vector Databases, Agent Workflows, Prompt Engineering

Backend & APIs: FastAPI, Flask, Spring Boot, Node.js, REST APIs

Databases & Cloud: PostgreSQL, MongoDB, ChromaDB, AWS (EC2, S3), Docker, Google Cloud Run

Core CS: Data Structures, Algorithms, OOP, System Design, Debugging, Testing

WORK EXPERIENCE

AI Engineer Intern | *Simple Ticket* | Boston, MA Jul 2025 – Sep 2025

- Developed AI-powered CRM assistants with OCR-based document ingestion, semantic retrieval, and **Retrieval-Augmented Generation (RAG)** pipelines using vector databases and LLMs.
- Built API-driven agent workflows using **LangGraph**, **LangChain Expression Language (LCEL)**, and **REST APIs** to support stateful multi-step task execution.
- Containerized AI services using **Docker** and deployed scalable inference workflows supporting low-latency response generation.
- Collaborated with product and engineering teams to deliver production-ready AI features and workflow automation solutions.

Research Assistant | *California State University, East Bay* | Hayward, CA Sep 2025 – Mar 2026

- Developed real-time robotic perception systems on **Jetson Orin Nano** using **ROS2**, **OpenCV**, **LiDAR**, and **depth cameras** for autonomous navigation and object tracking.
- Integrated **sensor fusion**, spatial mapping, and computer vision pipelines to improve navigation accuracy and reduce end-to-end latency by **15%**.
- Optimized multithreaded perception workflows and distributed **ROS2** communication pipelines to improve throughput and system responsiveness.

Firmware Engineer | *Indieseemic Private Limited* | Ahmedabad, India Jan 2024 – Jun 2024

- Developed embedded software on **Nordic nRF52/nRF53 DKs** and **Quectel EVKs** using **Zephyr RTOS**, **Nordic nRF Connect SDK**, and **Segger** debugging tools.
- Architected a **BLE** communication framework supporting **UART**, **SPI**, **I2C**, and **LoRa** interfaces, improving concurrent peripheral communication throughput by **10%**.

PROJECTS

Smart Recipe Tagger – Multimodal Recipe Platform [GitHub](#)

- Built a multimodal AI platform using **React**, **Node.js**, **MongoDB**, and **Firebase** for image understanding, recipe generation, and user content management.
- Designed inference pipelines integrating **Google Vision API** and **Gemini API** for image understanding and multimodal content generation.
- Deployed containerized services using **Docker**, **CI/CD**, and **Google Cloud Run** for scalable inference.

RAG-Based Personalized Outreach Engine [GitHub](#)

- Developed a retrieval-augmented generation platform using **Llama 3.3**, **LangChain**, **Groq**, and **ChromaDB** for personalized content generation.
- Built data-ingestion, embedding-generation, vector-indexing, and retrieval pipelines supporting low-latency semantic search.
- Exposed retrieval and generation workflows through REST-based services for interactive content generation.

FactShield – Ensemble Fake News Detection Platform [GitHub](#)

- Developed a **Flask**-based inference service integrating **RoBERTa**, **XGBoost**, **LightGBM**, and **Scikit-learn** for real-time misinformation classification.
- Designed an ensemble prediction pipeline combining **transformer embeddings** with traditional machine learning models to generate confidence-scored predictions.
- Built and evaluated ensemble models using cross-validation, achieving **93% F1-score**, **94% precision**, and **92% recall** on misinformation classification tasks.

Distributed Notification Platform [GitHub](#)

- Architected a distributed notification platform using **Apache Kafka**, **Spring Boot**, and microservices supporting **3 priority tiers** and **3 notification channels** (Email, SMS, Push) through asynchronous event-driven delivery pipelines.
- Implemented asynchronous event-driven workflows with retry handling, failure recovery, and external API orchestration for Email, SMS, and Push delivery.
- Designed event schemas and fault-tolerant consumer workflows to improve scalability, reliability, and failure isolation.